

Problem 1

Consider the Dihedral group D_{2n} and let k be a positive integer dividing n

1. prove that $\langle r^k \rangle$ is a normal subgroup of D_{2n}
2. Prove that $D_{2n}/\langle r^k \rangle \cong D_{2k}$

Solution:

let $ka = n$ for some $a \in \mathbb{Z}^+$

from previous knowledge, we know that $\langle r^k \rangle$ is a subgroup of D_{2n}

1. note that $\langle r^k \rangle = \{r^k, r^{2k}, \dots, r^{ak} = r^n = 1\} = \{r^{mk} : m \in 1, \dots, a\}$

Then $\langle r^k \rangle \trianglelefteq D_{2n} \Leftrightarrow dr^{mk}d^{-1} \in \langle r^k \rangle$ for any $d \in D_{2n}$

Let $d = e \in D_{2n}$ then clearly this is true

Let $d = r^j$ for some $j \in 1, \dots, n-1$ then $r^j r^{mk} r^{-j} = r^{j+mk-j} = r^{mk} \in \langle r^k \rangle$

Let $d = sr^j$ for some $j \in 1, \dots, n-1$ then

$$sr^j r^{mk} (sr^j)^{-1} = sr^j r^{mk} r^{-j} s = ssr^{-j} r^{-mk} r^j = r^{-mk} = (r^{mk})^{-1} \in \langle r^k \rangle$$

Let $d = s$ then $sr^{mk} s^{-1} = sr^{mk} s = ssr^{-mk} = r^{-mk} \in \langle r^k \rangle$ ■

2.

Write $D_{2n} = \langle r, s \mid r^n = s^2 = 1, sr = r^{-1}s \rangle$

Write $D_{2k} = \langle r', s' \mid r'^k = s'^2 = 1, s'r' = r'^{-1}s' \rangle$

By the definition of D_{2n} , let $x = r^a s^b$ and $y = r^c s^d$ where $0 \leq a, c < n$ and $0 \leq b, d < 2$

Define the homomorphism $\varphi : D_{2n} \rightarrow D_{2k}$ by $r^a s^b \mapsto r'^{a \bmod k} s'^b$

Then

$$\begin{aligned} \varphi(xy) &= \varphi(r^a s^b r^c s^d) = \varphi(r^a r^{-c} s^b s^d) = \varphi(r^{a-c} s^{b+d}) = r'^{a-c \bmod k} s'^{b+d} = r'^{a \bmod k} r'^{-c \bmod k} s'^b s'^d = \\ &= r'^{a \bmod k} s'^b r'^{c \bmod k} s'^d = \varphi(x)\varphi(y) \end{aligned}$$

so that φ is a homomorphism, and note that surjectivity follows immediately ($k \leq n$)

We can find the kernel of φ by the following: $\varphi(r^a s^b) = 1 \Leftrightarrow b = 0 \wedge a \equiv 0 \bmod k \Rightarrow \ker \varphi = \langle r^k \rangle$

Applying the first isomorphism theorem, we obtain $D_{2n}/\langle r^k \rangle \cong D_{2k}$

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Problem 2

Prove that if $G/Z(G)$ is cyclic then G is abelian. (If $G/Z(G)$ with generator $xZ(G)$, show that every element of G can be written in the form $x^a z$ for some $a \in \mathbb{Z}$ and some element $z \in Z(G)$)

Solution:

For some $x \in G$ let the generator of $G/Z(G)$ be $xZ(G)$ then we have that for $g \in G$ that $gZ(G) = (xZ(G))^a$ for some a and using the binary operation of $G/Z(G)$ we have that $(xZ(G))^a = x^a Z(G)$

Since we have that $gZ(G) = x^a Z(G)$ then by a proposition in class $(x^a)^{-1}g \in Z(G)$, so let $x^{-a}g = z$ for some $z \in Z$ then we have $g = x^a z$

Since z is from $Z(G)$ and commutes with all elements of G , let $g_1 = x^{a_1} z_1$ and $g_2 = x^{a_2} z_2$ then

$$g_1 g_2 = x^{a_1} z_1 x^{a_2} z_2 = x^{a_2} z_2 x^{a_1} z_1 = g_2 g_1$$

where $g_1, g_2 \in G$ and so G is abelian

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references:

<https://math.stackexchange.com/questions/63087/>

Problem 3

Show that if $|G| = pq$ for some primes p, q (not necessarily distinct), then either G is abelian or $Z(G) = \{1\}$

Solution:

by lagrange's theorem, the possible orders of $Z(G)$ are $1, p, q, pq$

Suppose that $|Z(G)| = 1$ then $Z(G) = \{1\}$ (trivial group is a group)

Suppose that $|Z(G)| = pq$ then $Z(G) = G$ so that G is abelian

Suppose that $|Z(G)| = p$ then $|G/Z(G)| = pq/p = q$ (from class) since q is prime $G/Z(G)$ is cyclic (from previous homework). This makes G abelian (from problem 2) i.e. that $Z(G) = G$ leading to contradiction $p = pq$ so we conclude $|Z(G)| \neq p$

Similarly, $|Z(G)| \neq q$

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Problem 4

Let G be a group, let N be a normal subgroup of G , and let $\overline{G} = G/N$. Prove that $\overline{x}$ and $\overline{y}$ commute in $\overline{G}$ if and only if $x^{-1}y^{-1}xy \in N$ (The element $x^{-1}y^{-1}xy$ is called the commutator of x and y and is denoted by $[x, y]$)

Solution:

Note that $xNyN = yNxN \Leftrightarrow xyN = yxN \Leftrightarrow (yx)^{-1}xy \in N \Leftrightarrow x^{-1}y^{-1}xy \in N$

By the binary operation in G/N and a proposition in class which stated that $aH = bH \Leftrightarrow a^{-1}b \in H$ or equivalently that $bH = aH \Leftrightarrow b^{-1}a \in H$

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Conversely, we can say that if

$$x^{-1}y^{-1}xy \in N \Rightarrow x^{-1}y^{-1}xyN = N \Rightarrow x^{-1}Ny^{-1}NxNyN = 1N \Rightarrow xNyN = yNxN$$

Problem 5

Let G be a group, prove that $N = \langle x^{-1}y^{-1}xy : x, y \in G \rangle$ is a normal subgroup of G and G/N is Abelian (Here N is called the commutator subgroup of G)

Solution:

1. $N \trianglelefteq G \Leftrightarrow g(x^{-1}y^{-1}xy)g^{-1} \in N$ for all $g \in G$ and $x^{-1}y^{-1}xy \in N$

Note that $gx^{-1}y^{-1}xyg^{-1} = (gx^{-1}g^{-1})(gy^{-1}g^{-1})(gxg^{-1})(gyg^{-1})(gg^{-1}) \in N$

since each of these terms are in G and that $(gxg^{-1})^{-1} = gx^{-1}g^{-1}$ and $(gyg^{-1})^{-1} = gy^{-1}g^{-1}$

2. let $g_1, g_2 \in G$ then $g_1^{-1}g_2^{-1}g_1g_2 \in N \Rightarrow (g_2g_1)^{-1}(g_1)(g_2) \in N \Rightarrow g_1g_2N = g_2g_1N \Rightarrow g_1Ng_2N = g_2Ng_1N$ (the second implication comes from the same proposition used in problem 4 from class)

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Problem 6

Prove that every subgroup of Q_8 is normal

Solution:

Q_8 and $\{1\}$ will be normal (from class)

$\{1, -1\}$ is normal since $x(-1)x^{-1} = -xx^{-1} = (-1)xx^{-1} = -1$ for any $x \in Q_8$

There are also 3 subgroups of order 4 of the form $\{1, -1, x, -x\}$ where $x \in \{i, j, k\}$

Note that i, j, k each have order 4, and that the conjugation gxg^{-1} by some $g \in Q_8$ must also have order 4

$$(\pm i)(\pm j) = \pm k$$

$$(\pm j)(\pm k) = \pm i$$

Due to the presentation of Q_8 , we necessarily have that $(\pm i)(\pm k) = \pm j$ (although Q_8 is non abelian, these properties hold no matter the order of multiplication)

Note that if $g = 1 \vee g = -1 \vee g = \pm x$ then gxg^{-1} will still be in the subgroup (where x is from) due to the presentation of Q_8 . So we need to look at the other cases of g :

So if we consider the subsets $\{\pm i\}, \{\pm j\}, \{\pm k\}$ and that $g \neq \pm x$, then g will be from one of these subsets, x another, so that gx will be in the third subset, and gxg^{-1} will be in the set that x is from since g^{-1} is in the same subset g is from

(i.e. i, j, k are “cyclically permuted” under the operation of Q_8)

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The definition of $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ has relations $i^2 = j^2 = k^2 = -1, ij = -ji = k, jk = -kj = i, ki = -ik = j, 1x = x1 = x, 1^2 = (-1)^2 = 1, (-1)x = x(-1) = -x$

Problem 7

Let G be the group $\mathbb{Z}_8 \times \mathbb{Z}_4$ and let $H = \langle (2, 2) \rangle$. You are given that G/H is isomorphic to one of $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, or $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. Determine which one by elimination.

Solution:

Note that $H = \{(0, 0), (2, 2), (4, 0), (6, 2)\}$

There are $|G| / |H| = 32/4 = 8$ cosets

Note that all non-identity elements of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ have order 2, while $\mathbb{Z}_8$ has an element of order 8

The equivalent cosets of G/H are all representatives plus a multiple of the generator $(2, 2)$

G/H has cosets with representatives

$(1, 0)$ with order 4

$(1, 1)$ with order 2

$(1, 2)$ with order 4

$(1, 3)$ with order 2

$(0, 0)$ with order 1

$(0, 1)$ with order 4

$(0, 2)$ with order 2

$(0, 3)$ with order 4

Therefore $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ cannot be the answer since it has no elements of order 4,

and since $\mathbb{Z}_8$ has an element $(1 \in \mathbb{Z}_8)$ which has order 8, then it cannot also be the answer

The only answer left is $\mathbb{Z}_4 \times \mathbb{Z}_2$ and based on the problem statement we can conclude that $G/H \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

Problem 8

Let G be a finite group and let H be a normal subgroup of G . Prove that the order of the element gH must divide the order of $g \in G$

Solution:

Note that $(gH)^{|g|} = g^{|g|}H = 1H = H$

Although we do not know the order of gH , by the division algorithm we can write $|g| = |gH|q + r$ for some $q, r \in \mathbb{Z}$ where $0 \leq r < |gH|$

Then $(gH)^{|g|} = (gH)^{|gH|q+r} = (gH)^{|gH|q}(gH)^r = (gH)^r$

This means that $H = (gH)^r$ but the above, $r < |gH|$ so we must have $r = 0$

Then $|g| = |gH|q \Rightarrow |gH| \mid |g|$

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<https://math.stackexchange.com/questions/1897118/>

Problem 9

Let C be a normal subgroup of the group A and D be a normal subgroup of the group B . Prove that $C \times D \trianglelefteq A \times B$ and $(A \times B)/(C \times D) \cong (A/C) \times (B/D)$

Solution:

1. take $(a, b) \in A \times B, (c, d) \in C \times D$ then $(a, b)(c, d)(a, b)^{-1} = (a, b)(c, d)(a^{-1}, b^{-1}) = (aca^{-1})(bdb^{-1}) \in C \times D$ since $C \trianglelefteq A \Rightarrow aca^{-1} \in C$ and $D \trianglelefteq B \Rightarrow bdb^{-1} \in D$

2. Define the homomorphism $\varphi : A \times B \rightarrow (A/C) \times (B/D)$ by $\varphi((a, b)) = (aC, bD)$

(φ is clearly surjective since any $(aC, bD) \in A/C \times B/D = \varphi(a, b), a \in A, b \in B$)

φ is a homomorphism since $\varphi((a, b)(c, d)) = \varphi((ac, bd)) = (acC, bdD) = (aC, bD)(cC, dD) = \varphi((a, b))\varphi((c, d))$ for some $(a, b) \in A \times B$ and $(c, d) \in A \times B$

the kernel of φ is the $(a, b) \in A \times B$ such that $\varphi((a, b)) = (aC, bD) = (1C, 1D) = (C, D)$.

Recall that $aC = C \Leftrightarrow a \in C, bD = D \Leftrightarrow b \in D$ from class, then the kernel of φ is exactly $C \times D$

By the first isomorphism theorem, we can conclude that $(A \times B)/\ker \varphi \cong \text{Im } \varphi$ i.e. $(A \times B)/(C \times D) \cong (A/C) \times (B/D) = \text{Im } \varphi$ (where the last equality holds due to the surjectivity of φ) ■